D. Traveling Graph

time limit per test: 0.5 second
memory limit per test: 64 megabytes

You are given undirected weighted graph. Find the length of the shortest cycle which starts from the vertex 1 and passes throught all the edges at least once. Graph may contain multiply edges between a pair of vertices and loops (edges from the vertex to itself).

Input

The first line of the input contains two integers n and m ($1 \le n \le 15$, $0 \le m \le 2000$), n is the amount of vertices, and m is the amount of edges. Following m lines contain edges as a triples x, y, w ($1 \le x, y \le n$, $1 \le w \le 10000$), x, y are edge endpoints, and w is the edge length.

Output

Output minimal cycle length or -1 if it doesn't exists.

Examples

input Copy

```
3 3
1 2 1
2 3 1
3 1 1
```

output Copy

```
3
```

input Copy

```
3 2
1 2 3
2 3 4
```

output Copy

```
14
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Alpha Round 21 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371474826	Apr/17/2026 12:14	Accepted
371472173	Apr/17/2026 11:47	Accepted
371471527	Apr/17/2026 11:40	Accepted

371471383	Apr/17/2026 11:38	Wrong answer on test 16
371471276	Apr/17/2026 11:37	Wrong answer on test 3

→ Problem tags

bitmasks graph matchings graphs *2400

No tag edit access

→ Contest materials

- Announcement

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